

Eleana Karachaliou

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Profile

PhD candidate at the University of Manitoba (Biological Sciences), expected to graduate Fall 2023 - Winter 2024. Background in evolutionary biology with a focus on the aquatic environment. I am currently working on anadromous sea lamprey population genomics, evolutionary history and implications on conservation and management. I am also working on a project on the environmental and human-caused drivers of genetic diversity in global marine fish species. I am interested in both the detail and greater picture of aquatic species' evolution, utilizing molecular methods, and combining different knowledge systems so as to assist conservation and management efforts while promoting scientific outreach and education.

Education

- 2022** (November) CCFFR - SCL Workshop on Indigenous Engagement (Online).
- 2020** (June/July) Online Workshop Series on Data Management and Synthesis in Ecology and Evolution – CIEE Living Data Project.
- 2019** (November) Genomic Solutions for Wildlife management/Conservation in Canada – Genome Prairie (Online).
- 2019** (October) Lamprey Genomics Workshop, Winnipeg, Canada.
- 2019 – Present** PhD, Department of Biological Sciences, Faculty of Science, University of Manitoba, Canada.
- 2017-2019** MSc Systems Biology, Department of Biotechnology, Agricultural University of Athens, Greece.
(Grade: 9.19/10)
- 2019** (February) Workshop: Molecular Evolution: Patterns and Causes – Konnevesi Research Station of the University of Jyväskylä, Finland.
- 2018** (June) Summer School in Aquatic Field Work, Technical University of Denmark (DTU).
- 2015** Erasmus exchange program -Spring Semester, BOKU, Vienna, Austria.
- 2011-2017** Integrated BSc and MSc in Ichthyology and Aquatic Environment, Department of Agriculture, Ichthyology and Aquatic Environment, School of Agricultural Sciences, University of Thessaly, Greece.
(Grade: 6.94/10)
- 2005-2011** Nikaia High School, Larisa, Greece.

Employment History

- 2021-2022** Teaching Assistant – Evolutionary Biology, University of Manitoba (Fall Semester).
- 2021-2022** Grader/Marker – Evolutionary Biology, University of Manitoba (Fall Semester).
- 2016-2017** English Teaching Assistant, Yellow Umbrellas Kindergarten, Athens, Greece.
Assisting the schools' English teacher with lessons and daily activities.
- 2015** Summer Internship, Multihull Yachting, Lavrio, Greece.
Assisting the organization of trips, maintenance and transport of vessels, communication with ports and customers.
- 2014** Summer Internship, Piraeus Directorate of Fisheries, Athens, Greece.
Responsible for issuing aquaculture and fishing permits, archives, and assisting professional training and outreach programmes for fishermen.

Research

- 2020 (Summer)** Field Assistant – Bee biodiversity monitoring across Manitoba.
- 2019 (Fall)** Field Assistant – Squirrel Microevolution in rapidly changing environments.
- 2019 - Present** Past & Contemporary Patterns of Marine Biodiversity: Making Inferences from Anadromous Sea Lamprey Population Genomics & Publicly Archived Genetic Data – PhD Thesis.
- 2018 - 2019** Biomonitoring of marine mammals in Eastern Mediterranean - Participation in research cruises.
Aim: Population Assessment for Dolphins of the Corinthian Gulf - Morphological identification of species and measurements for abundance and distribution.
- 2018 - 2019** DNA Barcoding and Phylogenetics of Elasmobranch Species in the Eastern Mediterranean - MSc thesis.
Aim: DNA Barcoding of the mitochondrial COI gene for species identification of different Eastern Mediterranean Elasmobranch species and determination of phylogenetic relationships.
- 2018** Assessment of Fish & Habitat in the Øresund- Project for Summer School in Aquatic Field Work, Technical University of Denmark.
Aim: To plan and carry out an experiment at sea and familiarization with basic experimental methodology - onboard fish sorting and use of rosette (CTD, Oxygen sensor, fluorescence sensor), otolith measurements and water sample chlorophyll analysis.
- 2018** Species Identification using Molecular Techniques for the Genus *Galeus* - Rotation at the Molecular Biology Laboratory (Agricultural University of Athens) as part of MSc Degree.
Aim: To use DNA Barcoding of the mitochondrial COI gene to assign specimens to the species *Galeus melastomus*, confirming morphological data.
- 2018** Location of *mosaic* mutation in At4g21400 and At4g21410 genes for *Arabidopsis Thaliana Columbia* ecotype - Rotation at the Molecular Biology Laboratory (Agricultural University of Athens) as part of MSc Degree.
Aim: To extend a previous study where the location of the mutation in the genome was calculated by mapping with respect to molecular markers which were polymorphic between two ecotypes. Using PCR and restriction enzymes, the mutation was located in the gene CRK28-At4g21400.
- 2015** Isolation & characterization of two *Hox* genes (A13a&A13b) in *Solea Solea* - Undergraduate thesis
Aim: The isolation and characterization of the *Hox* genes *Hox* A13a and *Hox* A13b, via PCR amplification of the genes of interest, for the common sole *Solea solea* which is a promising species for the future of aquaculture. Sequencing of the two *Hox* genes lead to the identification of expression, structure and differences between viable and non-viable egg batches.
- 2013-2015** Participation in a project on Marine Angiosperm Tolerance On Environmental Stress (Thales Project): Physiological, cellular, biochemical, molecular mechanisms related to species and habitat diversity.
Aim: Collecting marine angiosperms in the Pagasitikos and Sikia Gulf as part of a diving team. The species *Cymodocea nodosa*, *Halophila Stipulacea* and *Posidonia oceanica* were collected and observations made on their occurrence and distribution.

Technical Skills

- Molecular biology (PCR, DNA extraction).
- Programming languages: R, bash command line (Linux/Unix), working with computer clusters.
- Processing and interpreting whole-genome sequencing data (e.g. SNP calling, Sequentially Markovian Coalescent Methods).
- GIS, ClustalX, MEGA7, Logger 2010 and PAMGUARD software, Microsoft Office.

- Experimental and field work planning and the organizational aspect of working on research vessels.
- Scientific Diving.
- Taxonomy and ecology of benthic invertebrates (Project during Erasmus exchange at BOKU Vienna).
- Experience in marine mammal observation and passive acoustic monitoring.

Languages

- Bilingual English and Greek
- Anishnaabemowin/Ojibwe - Beginners level

Qualifications

- IELTS English Certificate
- PADI Open Water Diving Certificate

Awards

- ICES PICES Early Career Scientist Conference 2022 Best Poster Award (2022).
- MEOPAR Conference Travel Award (2022).
- Department of Biological Sciences & Faculty of Science Travel Award, University of Manitoba (2022)
- International Graduate Student Entrance Scholarship, University of Manitoba (2019)

Conference Presentations

- **Karachaliou, E.**, C. Schmidt, M.F. Docker, C.J. Garroway (2022) Environmental & human disturbance effects on global marine genetic diversity. ESA & CSEE Joint Meeting, August 2022.
- **Karachaliou, E.**, C.J. Garroway, P. Grayson, G. Evanno, M.F. Docker (2022) Whole genome re-sequencing uncovers historic transoceanic divergence in anadromous sea lamprey *Petromyzon marinus* – implications for conservation & management. Fourth ICES PICES Early Career Scientist Conference, (Poster) July 2022.
- Suurvali, J., P. Grayson, M.L. Mahoney, J.L. Ogden, **E. Karachaliou**, M.J. Lawrence, K.M. Jeffries, M.F. Docker, C.J. Garroway (2022) Genomic signatures of adaptation in invasive sea lamprey of the Laurentian Great Lakes. ESA & CSEE Joint Meeting, (Oral) August 2022.
- **Karachaliou, E.**, C. Schmidt, M.F. Docker, C.J. Garroway (2021) Global drivers of genetic diversity in marine fish species. CSEE Virtual Meeting, August 2021.
- **Karachaliou, E.**, C. Garroway, P. Grayson, G. Evanno, and M.F. Docker (2021) Reconstructing anadromous sea lamprey evolutionary history – implications for management & conservation. CCFFR-SCL Virtual Meeting, February 2021.
- **Karachaliou E.**, P. Hatzopoulos, A. Exadactylos, G.A. Gkafas, P. Megalofonou (2018) Cytochrome Oxidase Subunit I (COI) Gene - A Method for Species Identification of the Blackmouth Catshark (*Galeus melastomus*). HydroMedit 2018 3rd International Congress on Applied Ichthyology & Aquatic Environment, November 2018.

Other

- Poster Session Judge, MEOPAR Annual Networking Meeting, Online (November 2022).
- Member of Search Committee for the position of Assistant Professor, University of Manitoba (Current).
- Honour the Wykanish Event (August 2022): Participation in community outreach event with Yakama Nation Fisheries, Prescott, WA, USA.
- Seminar presentation as part of CRITFC Brown Bag Seminar Series (August 2022) Title: Implications of marine genetic diversity patterns on conservation & management: what can we learn from anadromous sea lamprey & publicly archived genetic data? Portland, OR, USA.
- Research Visit - Columbia River Inter-Tribal Fish Commission & Yakama Nation Fisheries, Portland OR & Toppenish WA, USA (Summer 2022). Field work with Pacific lamprey, fin clips for DNA sampling, electrofishing, fish handling, tagging, habitat surveys, meeting with genetic monitoring experts.
- Guest speaker at Indigenous Engagement Session - Mississauga First Nation, Online (June 2022). An information, engagement session on lampricide treatments and concerns in the Laurentian Great Lakes.
- Session Moderator, CSEE-SCEE Online Conference (August 2021).
- Presenter at Sea Lamprey Webinar - County Administrative Board of Halland County, Swedish Agency for Marine and Water Management and SLU Swedish Species Information Centre, Online (April 2021).
- English language narrator for short documentary produced as part of the project “Biomonitoring of marine

- mammals in the Eastern Mediterranean” (2019).
- Interpreter at the 1st Hydromedit International Congress 2014 for the session: Round Table on Management Difficulties of Lake Karla.
- Participation in the 20th European Youth Parliament National Selection Conference 20-22 November 2009, Thessaloniki, Greece.

Hobbies

- Martial Arts (Muay Thai), Hiking, Diving.

Referees

- Margaret F. Docker Professor, Department of Biological Sciences, University of Manitoba, Winnipeg, Canada (PhD Supervisor) email: Margaret.Docker@umanitoba.ca
- Colin J. Garroway, Associate Professor, Department of Biological Sciences, University of Manitoba, Winnipeg, Canada (PhD co- Supervisor) email: Colin.Garroway@umanitoba.ca
- Stephen Petersen, Director, Conservation and Research, Assiniboine Park Zoo, Winnipeg, Canada (PhD Committee Member) email: spetersen@assiniboinepark.ca